

NEXURA TECH TEAM RECRUITMENT

Round 3 ~ Task Sheet

Language Task 1 – Student Management System

Using C++ create a basic student management system with add, display, search, update, and delete operations. Store name, enrollment number, branch, semester, and marks.

C++ Program: Student Management System

```
1 #include <iostream>
2 using namespace std;
3
4 struct Student {
5     string name, enrollment, branch;
6     int semester;
7     float marks;
8 };
9
10 Student s[100];
11 int n = 0;
12
13 void addStudent() {
14     cout << "\nEnter Name: ";
15     cin >> s[n].name;
16
17     cout << "Enter Enrollment Number: ";
18     cin >> s[n].enrollment;
19
20     cout << "Enter Branch: ";
21     cin >> s[n].branch;
22
23     cout << "Enter Semester: ";
24     cin >> s[n].semester;
25
26     cout << "Enter Marks: ";
27     cin >> s[n].marks;
28
29     n++;
30     cout << "Student added successfully!\n";
31 }
```

```
33 void displayStudents() {
34     if (n == 0) {
35         cout << "No students found.\n";
36         return;
37     }
38
39     for (int i = 0; i < n; i++) {
40         cout << "\nName: " << s[i].name;
41         cout << "\nEnrollment: " << s[i].enrollment;
42         cout << "\nBranch: " << s[i].branch;
43         cout << "\nSemester: " << s[i].semester;
44         cout << "\nMarks: " << s[i].marks << "\n";
45     }
46
47     void searchStudent() {
48         string enroll;
49         cout << "Enter Enrollment Number: ";
50         cin >> enroll;
51
52         for (int i = 0; i < n; i++) {
53             if (s[i].enrollment == enroll) {
54                 cout << "\nName: " << s[i].name;
55                 cout << "\nBranch: " << s[i].branch;
56                 cout << "\nSemester: " << s[i].semester;
57                 cout << "\nMarks: " << s[i].marks << "\n";
58                 return;
59             }
60         }
61     }
62 }
```

```
63 cout << "Student not found.\n";
64 }
65
66 void updateStudent() {
67     string enroll;
68     cout << "Enter Enrollment Number: ";
69     cin >> enroll;
70
71     for (int i = 0; i < n; i++) {
72         if (s[i].enrollment == enroll) {
73             cout << "Enter New Marks: ";
74             cin >> s[i].marks;
75
76             cout << "Student updated successfully!\n";
77             return;
78         }
79     }
80
81     cout << "Student not found.\n";
82 }
83
84 void deleteStudent() {
85     string enroll;
86     cout << "Enter Enrollment Number: ";
87     cin >> enroll;
88
89     for (int i = 0; i < n; i++) {
90         if (s[i].enrollment == enroll) {
91             for (int j = i; j < n - 1; j++)
92                 s[j] = s[j + 1];
93             n--;
94             cout << "Student deleted successfully!\n";
95             return;
96         }
97     }
98
99     cout << "Student not found.\n";
100 }
101
102 int main() {
103     int choice;
104
105     do {
106         cout << "\n\n-- Student Management System --";
107         cout << "\n1. Add Student";
108         cout << "\n2. Display Students";
109         cout << "\n3. Search Student";
110         cout << "\n4. Update Student";
111         cout << "\n5. Delete Student";
112         cout << "\n6. Exit";
113         cout << "\nEnter your choice: ";
114         cin >> choice;
115
116         switch (choice) {
117             case 1: addStudent(); break;
118             case 2: displayStudents(); break;
119             case 3: searchStudent(); break;
120             case 4: updateStudent(); break;
121             case 5: deleteStudent(); break;
122             case 6: cout << "Exiting..."; break;
123             default: cout << "Invalid choice!";
124         }
125     } while (choice != 6);
126
127     return 0;
128 }
```

```
129 }
```

```
129 }
```


OUTPUT:-

```
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit

Enter your choice: 1

Enter Student Name: Akshat
Enter Enrollment Number: 101
Enter Branch: CSE
Enter Semester: 1
Enter Marks: 85

Student added successfully!

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit

Enter your choice: 2

----- Student Records -----

Student 1
Name: Akshat
Enrollment No.: 101
Branch: CSE
Semester: 1
Marks: 85

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit

Enter your choice: 3

Enter Enrollment Number to search: 101

Student Found!
Name: Akshat
Enrollment No.: 101
Branch: CSE
Semester: 1
Marks: 85

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit

Enter your choice: 3

Enter Enrollment Number to search: 101

Student Found!
Name: Akshat
Enrollment No.: 101
Branch: CSE
Semester: 1
Marks: 85

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit

Enter your choice: 3

Enter Enrollment Number to search: 101

Student Found!
Name: Akshat
Enrollment No.: 101
Branch: CSE
Semester: 1
Marks: 85

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit

Enter your choice: 4

Enter Enrollment Number to update: 101
Enter New Name: Ansh
Enter New Branch: ECE
Enter New Semester: 1
Enter New Marks: 70

Student record updated successfully!

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit

Enter your choice: 5

Enter Enrollment Number to delete: 101

Student record deleted successfully!

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit

Enter your choice: 6

Exiting program...

Program finishedj
```